

Response: the CSL/SCG open issue is not open

The note is a faithful rendering of both what was said in discussion and what `PROOFS_v2.tex` currently states. Both are out of date on this point, and the note inherits the staleness in a way that matters: it presents as an unresolved question something that has already been computed, and it proposes a route to closing it that is already known to fail. What follows corrects each in turn and states what should replace them.

The estimated ratio has been computed. The note says that whether the estimated variance ratio is invariant to the preference parameter “has not been established.” It has. `lambda_RN_measured.py` simulates the solved model, samples it quarterly over a sixty-quarter window after a long burn-in, and applies the panel’s own estimator—the same estimator that would be used on real data—at four hundred paths per value of the asymmetry parameter. Under the first cost convention the median estimated ratio runs from approximately 1.334 at the neutral parameter value down to approximately 1.152 at the top of the admissible range; under the second it runs from approximately 1.390 to 1.167. The estimated ratio therefore *does* vary with the preference parameter, it decreases, and the direction is known.

The sentence in `PROOFS_v2.tex` asserting that this “is not known” was already flagged for revision before this note was written. It was never edited, and the note has now carried it into a second document. That is precisely how a stale line becomes a stale literature, and it is the strongest practical argument for making the edit rather than continuing to work around it.

The proposed closure route is already known to fail. The note proposes closing the gap by showing that the estimated ratio “converges to or otherwise tracks the definitional one as sample size grows.” That cannot be shown, because it is false. The panel estimator is asymptotically biased toward one: a true value of 1.30 measures 1.164 at sixty observations, 1.145 at five hundred, and 1.140 at twenty thousand. The bias converges to roughly -0.16 and stays there. This is not a finite-sample deficiency that more data repairs; the estimator targets a *different estimand*, because states near the classification boundary are assigned to one regime while their innovations reflect a mixture of both. The bridge the note asks for does not exist and will not be built. Treating this as pending an asymptotic argument points at a question that has already returned a negative answer.

What should replace the open-issue framing. The correct statement is not that volatility is uninformative and the estimated case is unchecked. It is a pair of claims at different strengths. First, the definitional evaluation of the ratio carries no information about the preference whatsoever—this is proved (SCG), and it is a derived consequence of how the cap’s geometry propagates into the saturation limits, not a definitional identity. Second, the estimated ratio carries a weak signal, and the weakness is quantifiable: roughly 0.18 of total movement across the entire admissible range of the preference parameter, against a per-path standard deviation of 0.16 to 0.20. For a single institution the signal is indistinguishable from its own noise; only a median across many institutions is informative at all.

That is a quantitative weak-identification claim rather than a blank uninformativeness claim, and it is the stronger position of the two. It survives the obvious objection—someone will estimate the ratio, obtain a number that moves with the parameter, and the paper needs to have predicted that movement and characterised its size rather than been surprised by it.

Consequence for the choice of first empirical test. The note’s conclusion is right and its reasoning is not. The K-sweep prediction (KSW) is the correct first test, but not because the definitional-versus-estimated relationship is unresolved groundwork. It is correct because that relationship *is* resolved, and unfavourably: the estimated ratio is a biased, noisy, weakly-responsive proxy for a quantity that is provably constant. A threshold-based test therefore dominates a variance-based one on identification grounds, which is a positive argument for KSW rather than a deferral of the alternative.

Two smaller points. The phrase “true by construction rather than by measurement” undersells SCG and invites a referee to read it as tautological; it is a proved symbolic result, and “proved from the model’s structure” carries the same meaning without the concession. And the suggestion that closure would proceed “presumably through simulation” describes work already performed at four hundred paths per parameter value, which will read oddly to anyone who has seen the measured-scale computation.

Shortest path to fixing this. One edit to the open-questions section of `PROOFS_v2.tex`, replacing the “is not known” sentence with the measured direction and the weak-identification magnitude, followed by a rewrite of this note against the corrected text. The note’s structure survives; only its premise changes.